Yash Shukla

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Professional Summary

Full Stack AI Engineer with 3.8 years of experience architecting high-performance web platforms and production-ready Generative AI solutions. Proven track record in engineering scalable, event-driven backends using Node.js, TypeScript, and Python (FastAPI), paired with modern responsive interfaces in React.js and Next.js.

Specialized in building end-to-end GenAI architectures—including advanced RAG pipelines (ChromaDB, LangChain, Cross-Encoder Reranking, Hybrid Search), multimodal LLM workflows, and asynchronous processing using BullMQ, Redis, and WebSockets. Adept at bridging robust full-stack software design with modern AI capabilities to deliver enterprise-grade applications from prototype to scale.

Live Projects

- **ScalePart (eVehicle Platform):** https://scalepart.com
- **Jalshaala (MOOC Ed-Tech Platform):** https://new.jalshaala.com

Experience

Troology Sept 2025 – Present

Full Stack AI Developer *On-Site*

- Currently spearheading the **eVehicle platform (ScalePart)**, developing both frontend architectures and backend AI integrations to handle unstructured catalog data.
- Engineered an **AI-driven document extraction pipeline** utilizing external LLM endpoints to automatically parse complex PDF manuals and unstructured text, extracting precise product model numbers, series names, and technical specifications.
- Built an **intelligent caching and confidence scoring layer** for AI responses, significantly reducing external API latency and costs while ensuring highly accurate data mapping to MongoDB schemas.
- Contributed to the **UPSDMA (Uttar Pradesh State Disaster Management Authority)** platform (**Rahat Sewa App**) for SDRF personnel, building features ensuring highly concurrent usage and operational reliability.
- Actively exploring and upskilling in **Google Cloud Platform (GCP)** and its suite of ML services to architect scalable AI infrastructures and predictive data models.
- Developing and optimizing **RESTful** architectures with **Node.js, Express.js, TypeScript**, and ensuring seamless state integration in **React.js** applications.

Cognitivo Jun 2024 – Aug 2025

Full Stack Developer *Hybrid*

- Integrated **AI-powered conversational features (OpenAI GPT-based)** directly into applications, providing intelligent user assistance, automating responses, and increasing user retention.
- Architected and optimized end-to-end **RESTful and GraphQL APIs** using **Node.js** and **PostgreSQL/MongoDB**, improving query efficiency and supporting a **50% increase in concurrent user traffic**.
- Built **Web Push Notifications** using Service Workers and VAPID keys to implement real-time event-driven alert systems.
- Integrated complex **third-party platforms** including Monday.com via custom webhooks and automation scripts, accelerating cross-team operational efficiency.
- Refactored legacy UI components using **React.js, TypeScript, Material UI, and Tailwind CSS**, drastically improving frontend performance, responsiveness, and web accessibility.

Singularies Pvt Ltd Jan 2023 – May 2024

Software Engineer *Remote*

- Designed and implemented robust features for a key product, writing maintainable logic utilizing **TypeScript, React.js, and Remix** frameworks.
- Developed scalable backend microservices using **Go (Golang)** and integrated critical third-party APIs ensuring high reliability and fault tolerance.
- Created high-performance, reusable UI components with **Tailwind CSS**, focusing on an inclusive, accessible user experience across all devices.

Projects

OmniRAG (Multimodal RAG & DB Agent) | *FastAPI, React, LangChain, ChromaDB, Redis* **Source Code**

- Architected a scalable full-stack Generative AI platform integrating a **FastAPI AI microservice**, **Node.js backend**, and **React** client for real-time document intelligence.
- Engineered an advanced RAG pipeline with **ChromaDB**, **Cross-Encoder Reranking**, and an **Agentic Web Search fallback** (DuckDuckGo) to resolve missing context.
- Implemented an autonomous **Text-to-MQL Agent** that translates natural language prompts into MongoDB aggregation pipelines to query and summarize live database records.
- Built an asynchronous document ingestion engine using **BullMQ and Redis**, decoupling PDF parsing from the main event loop with real-time **WebSocket** updates.
- Delivered real-time LLM token streaming via Server-Sent Events (SSE) using Groq-accelerated Llama-3.1, cutting user-perceived latency by 65%.

ScalePart (eVehicle Platform) | *React, Node.js, AI Integration, MongoDB* **Live Link**

- Developed and architected an end-to-end intelligent eVehicle platform, delivering robust backend APIs and a highly responsive React frontend.
- Engineered a seamless AI-driven data extraction pipeline that parses complex PDF documents and unstructured text into structured database records.
- Implemented an intelligent caching and confidence-scoring system to optimize AI queries, drastically reducing external LLM API latency and operational costs.
- Designed and integrated scalable MongoDB schemas for product catalogues, ensuring highly efficient data retrieval and optimal user experience.

Twitter Clone | *Node.js, Express, MongoDB, JWT* **Source Code**

- Developed a robust backend architecture using **Node.js and Express.js**, providing core functionalities for user authentication, tweet creation, and dynamic timeline management.
- Implemented secure authentication flows utilizing **JSON Web Tokens (JWT)** and password hashing with **bcryptjs** to protect user data and sessions.
- Integrated complex relational features including user associations (follow/unfollow) and interaction systems (likes and comments), ensuring efficient data management in **MongoDB**.
- Designed and rigorously tested a comprehensive **RESTful API** using **Postman**, ensuring reliable, well-documented endpoints for seamless frontend integration.

Technical Skills

Generative AI & ML: RAG Architecture, LangChain, Vector Databases (ChromaDB), HuggingFace, Cross-Encoder Reranking, Prompt Engineering, Autonomous Agents, LLM APIs (Groq, OpenAI, Vertex AI)

Languages: Python, TypeScript, JavaScript, SQL, Go, C++, HTML/CSS

Backend & Distributed: Node.js, Express.js, FastAPI, BullMQ, Redis, WebSockets (Socket.io), REST & GraphQL APIs

Frontend: React.js, Next.js, Remix, Tailwind CSS, Redux Toolkit, Material UI

Databases & Cloud: MongoDB, PostgreSQL, Google Cloud Platform (GCP), ChromaDB, Docker, Git, Postman

Achievements

- Ranked **1st in College** on GeeksforGeeks
- **3 Star Coder** on **LeetCode** (1540 rating, Top 34%)
- Solved **1000+ Data Structure and Algorithm Problems** across LeetCode, GFG, and CodeChef
- Maintained an **831-day longest streak on LeetCode**

Education

Rajkiya Engineering College Bijnor 2020 – 2024
Bachelor of Technology in Electrical Engineering; CGPA: 7.56/10 *Bijnor, India*